

C A S E S T U D Y · 2 0 2 6

EuFund

Private-source funding workflow; public exhibit case study

Status & evidence boundary — What this case study does and does not claim

Status: EuFund is a private-source portfolio project in active development; it is not presented as open source. A public exhibit extracted from the platform is published at github.com/Bogdan0708/eufund-exhibit (57 tests), covering the agent pipeline, retrieval and routing layers described below. This case study describes the portfolio implementation and design intent; it is not evidence of a production service, regulatory certification, customer usage or autonomous submission. No public EuFund service is claimed.

Role: Founder and sole engineer; domain input from prior EU-funded project delivery.

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01 Problem & workflow

Human control around funding work

Problem

Funding discovery and applications combine fragmented programme material, eligibility rules, narrative drafting and document review. The design focuses on making those steps explicit and reviewable.

Workflow design

- Phase-gated agent pipeline: discovery → research → structuring → drafting → review, each phase bound by a tool allowlist and a write rollout gate.
- Cited evidence retrieval over programme material (Qdrant) with citation validation before a claim is used.
- Tiered multi-provider routing (Claude / OpenAI / Gemini / Perplexity) with cost tables and cross-provider failover.
- Structured outputs with one re-prompt on schema failure; users remain responsible for submission.

02 Architecture

Implementation described conservatively

Primary stack

The reviewed private source is centred on Next.js and TypeScript, with PostgreSQL/Drizzle behind user-scoped row-level security, Qdrant, Redis, structured validation and configurable LLM providers. Python tooling supports offline agent/evaluation work rather than defining the main web runtime.

Deployment reference: a Cloud Build → Cloud Run pipeline with a pre-deploy Cloud SQL backup step, a migration job, a deploy step and a post-deploy health gate.

Engineering choices

- Workflow state controls sequence; model calls supply bounded reasoning or drafting inside phase and tool-allowlist limits.
- Eligibility and factual outputs are designed to retain source citations, validated before use.
- A failed structured-output schema check triggers one re-prompt rather than a silent fallback.
- Provider routing and retrieval are implementation surfaces that still require environment-specific validation.

03 Outcome & next steps

What the project proves today

Portfolio evidence

The public exhibit (github.com/Bogdan0708/eufund-exhibit, 57 tests) and the private source together demonstrate domain modelling, phase-gated agent orchestration, cited retrieval, multi-provider routing with failover, and a security-conscious, row-level-secured data design. These are implementation signals, not proof of live operation.

Next evidence gates

- Run and publish repeatable test/evaluation evidence without exposing private applicant data.
- Validate citations and eligibility against a dated corpus snapshot.
- Complete security review and a production health check before making hosting claims.